

Figure 1: Distributed network monitoring

Protocols

With the introduction of carrier-grade Ethernet, the problem management lifecycle as captured by OAMPT (operations, administration, maintenance, provisioning and troubleshooting) has gained importance. The telecommunication management network model has FCAPS (fault management, configuration, accounting/administration, performance and security) capabilities. Since service delivery and service support is what the end users focus on, ITIL (Information Technology Infrastructure Library) captures the workflows.

CMIP (Communication Management Information Protocol) is a network management protocol standardised by the ITU (International Telecommunication Union). CMIP models management information in terms of managed objects. With the establishment of the IETF (Internet Engineering Task Force), a much more simplified yet comprehensive ASN.1 based network management protocol that could be used to design a hierarchy of managed objects was created. This is the Simple Network Management Protocol (SNMP).

Managed objects

In a network, a managed object is a device (router, switch), a system (server, blade chassis), a service (PostgreSQL, Squid or HA-proxy), an entity (it could even be virtual machine or Docker instance) or a QoS (quality of service) metric that requires some form of monitoring and management.

A managed object is an abstract representation of network resources that are managed. It is defined in terms of the attributes it possesses, the operations that can be performed on it, and its relationship with other managed objects.

SNMP architecture

Simple Network Management Protocol (SNMP) is an IETF (Internet Engineering Task Force) ratified (multiple RFCs) standard for the Internet Management Protocol.

The notion of SNMP is expressed in terms of two complementary approaches:

1. The SNMP model of how to access the same information (communication over UDP and BER for encoding).
2. The SMI (structure of management information), which is a data model for representing a managed object as a MIB (management information base).

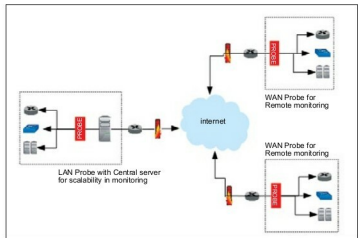


Figure 2: Probe based network monitoring approach

SNMP verbs

The basic verbs of the SNMP protocol are GET, SET, GETNEXT and GETBULK. An SNMP entity can also send asynchronous events, such as a TRAP and a NOTIFICATION. The verbs like GET and SET operate on individual variables of simple types, which can be scalar or located in tables.

Each SNMP data entity is expressed in terms of an OBJECT-IDENTIFIER encoded using TLV and a subset of ASN.1, which represents a node in a global tree. All SNMP definitions - MIBs, objects, etc - are identified by their OID in the global tree.

Thus, as part of the network monitoring strategy, it is common to define SMI for each of the network elements that need to be monitored and measured. An illustrative MIB fragment from mibdepot.com for D-Link is shown below.

D-Link: DLINKMGMT-MIB (SMIV2)

```
agentIpAddress OBJECT-TYPE
-- 1.3.6.1.4.1.326.2.20.1.11.1.3.2.1.2
-- iso(1). org(3). dod(6). internet(1). private(4).
enterprises(1). marconi(326). systems(2). external(20).
dlink(1). dlinkMgmtMIB(11). agentConfigInfo(1).
agentIpProtoConfig(3). agentIpIFTable(2). agentIpIFEntry(1).
agentIpAddress(2)
SYNTAX IpAddress
MAX-ACCESS read-only
DESCRIPTION
"The Ip address of the interface ."
 ::= { agentIpIFEntry 2 }
```

JMX architecture

Java Management Extensions (JMX) is a framework and technology platform for building distributed and modular solutions to manage and monitor devices, applications and service-driven networks. Legacy systems can be adapted with the JMX infrastructure. While it is a Java based technology, JMX Beans and mbean server can be used to manage complex network topologies.

OpenNMS is an open source project that offers a feature-rich network management application platform.